Vulnerability Assessment & Penetration Testing (VAPT)

WEEK 2 – Vulnerability Assessment & Documentation Report

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Environment: Host-Only / Internal Network (Isolated Virtual Lab)

Tools Used: Nmap, OpenVAS (Greenbone Vulnerability Manager)

1. Objective

The objective of Week 2 was to conduct detailed network scanning, service enumeration, and automated vulnerability assessment on the identified target system to identify security weaknesses and classify associated risks.

2. Network Discovery & Port Scanning

Network discovery was performed to identify active hosts within the lab subnet. Further enumeration detected open ports, services, and operating system details.

Commands Executed:

nmap -sn 192.168.230.0/24

nmap -sV -O 192.168.230.131

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┌──(root㉿kali)-[/home/divakar]
└─# nmap -sV -O -p- -sS 192.168.230.131
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-27 14:38 IST
Stats: 0:01:35 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 14:40 (0:00:03 remaining)
Nmap scan report for 192.168.230.131
Host is up (0.0013s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rexecd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
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2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
41619/tcp open  java-rmi    GNU Classpath grmiregistry
42642/tcp open  status      1 (RPC #100024)
51997/tcp open  nlockmgr    1-4 (RPC #100021)
54670/tcp open  mountd      1-3 (RPC #100005)
MAC Address: 00:0C:29:AD:2E:80 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.85 seconds
```

3. Service Enumeration Results

Major services identified:

- FTP – vsftpd 2.3.4
- SSH – OpenSSH 4.7p1

• Telnet – Unencrypted service

• HTTP – Apache 2.2.8

• SMB – Samba 3.x

• MySQL – 5.0.51a

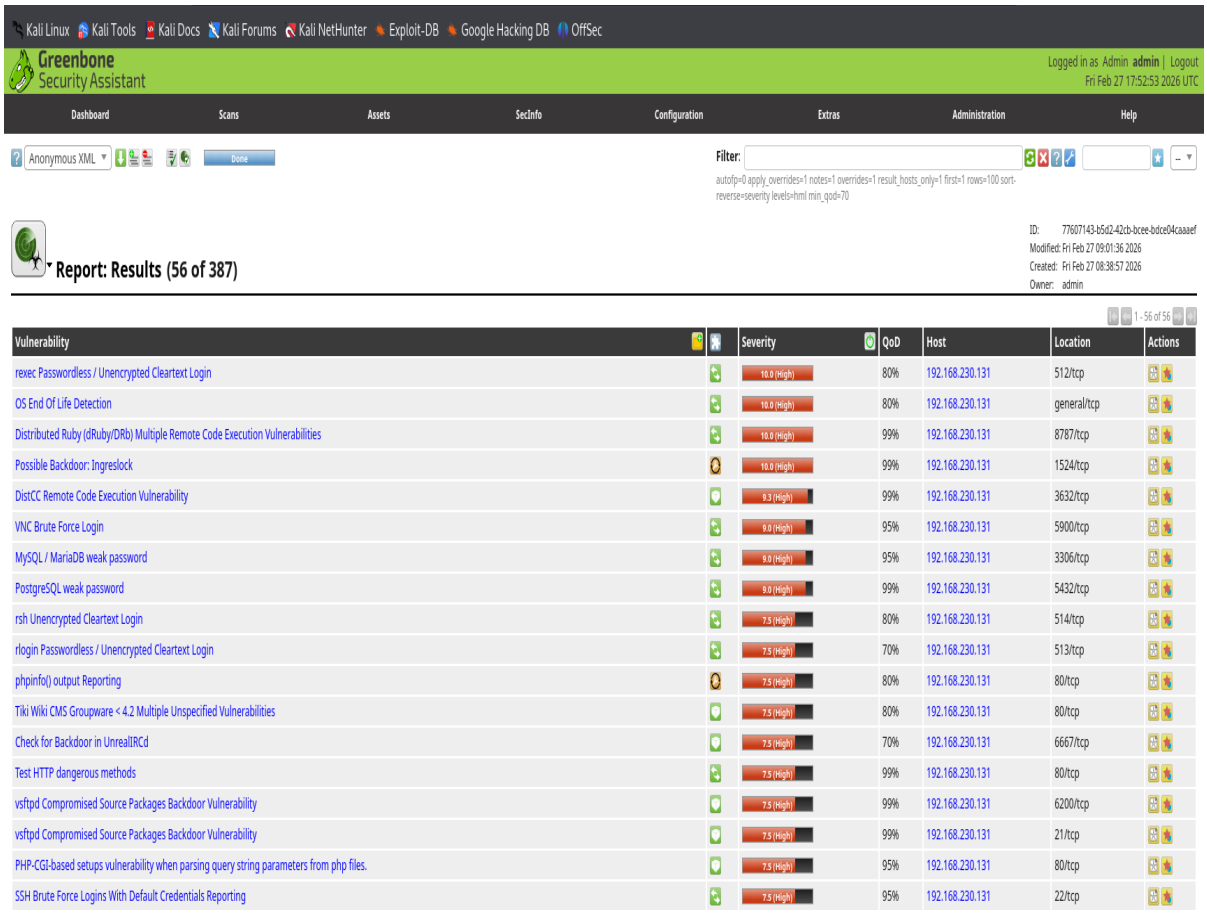
Outdated services significantly increased the attack surface of the system.

4. Vulnerability Assessment Using OpenVAS

OpenVAS (Greenbone Vulnerability Manager) was used to perform automated vulnerability scanning using the 'Full and Fast' scan profile.

Target IP: 192.168.230.131

Scan Profile: Full and Fast



Vulnerability	Severity	QoD	Host	Location	Actions
rexec Passwordless / Unencrypted Cleartext Login	10.0 (High)	80%	192.168.230.131	512/tcp	
OS End Of Life Detection	10.0 (High)	80%	192.168.230.131	general/tcp	
Distributed Ruby (dRuby/DRb) Multiple Remote Code Execution Vulnerabilities	10.0 (High)	99%	192.168.230.131	8787/tcp	
Possible Backdoor: Ingreslock	10.0 (High)	99%	192.168.230.131	1524/tcp	
DistCC Remote Code Execution Vulnerability	9.3 (High)	99%	192.168.230.131	3632/tcp	
VNC Brute Force Login	9.0 (High)	95%	192.168.230.131	5900/tcp	
MySQL / MariaDB weak password	9.0 (High)	95%	192.168.230.131	3306/tcp	
PostgreSQL weak password	9.0 (High)	99%	192.168.230.131	5432/tcp	
rsh Unencrypted Cleartext Login	7.5 (High)	80%	192.168.230.131	514/tcp	
rlogin Passwordless / Unencrypted Cleartext Login	7.5 (High)	70%	192.168.230.131	513/tcp	
phpinfo() output Reporting	7.5 (High)	80%	192.168.230.131	80/tcp	
Tiki Wiki CMS Groupware < 4.2 Multiple Unspecified Vulnerabilities	7.5 (High)	80%	192.168.230.131	80/tcp	
Check for Backdoor in UnrealIRCd	7.5 (High)	70%	192.168.230.131	6667/tcp	
Test HTTP dangerous methods	7.5 (High)	99%	192.168.230.131	80/tcp	
vsftpd Compromised Source Packages Backdoor Vulnerability	7.5 (High)	99%	192.168.230.131	6200/tcp	
vsftpd Compromised Source Packages Backdoor Vulnerability	7.5 (High)	99%	192.168.230.131	21/tcp	
PHP-CGI-based setups vulnerability when parsing query string parameters from php files.	7.5 (High)	95%	192.168.230.131	80/tcp	
SSH Brute Force Logins With Default Credentials Reporting	7.5 (High)	95%	192.168.230.131	22/tcp	

UnrealIRCd Authentication Spoofing Vulnerability		6.8 (Medium)	80%	192.168.230.131	6667/tcp
TWiki Cross-Site Request Forgery Vulnerability - Sep10		6.8 (Medium)	80%	192.168.230.131	80/tcp
Multiple Vendors STARTTLS Implementation Plaintext Arbitrary Command Injection Vulnerability		6.8 (Medium)	99%	192.168.230.131	25/tcp
SSL/TLS: OpenSSL CCS Man in the Middle Security Bypass Vulnerability		6.8 (Medium)	70%	192.168.230.131	5432/tcp
Tiki Wiki CMS Groupware < 17.2 SQL Injection Vulnerability		6.5 (Medium)	80%	192.168.230.131	80/tcp
Anonymous FTP Login Reporting		6.4 (Medium)	80%	192.168.230.131	21/tcp
TWiki Cross-Site Request Forgery Vulnerability		6.0 (Medium)	80%	192.168.230.131	80/tcp
HTTP Debugging Methods (TRACE/TRACK) Enabled		5.8 (Medium)	99%	192.168.230.131	80/tcp
Check if Mailserver answer to VRFY and EXPN requests		5.0 (Medium)	99%	192.168.230.131	25/tcp
Tiki Wiki CMS Groupware Input Sanitation Weakness Vulnerability		5.0 (Medium)	80%	192.168.230.131	80/tcp
/doc directory browsable		5.0 (Medium)	80%	192.168.230.131	80/tcp
Tiki Wiki CMS Groupware 'fixedURLData' Local File Inclusion Vulnerability		5.0 (Medium)	80%	192.168.230.131	80/tcp
SSL/TLS: Certificate Expired		5.0 (Medium)	99%	192.168.230.131	25/tcp
SSL/TLS: Certificate Expired		5.0 (Medium)	99%	192.168.230.131	5432/tcp
awiki Multiple Local File Include Vulnerabilities		5.0 (Medium)	99%	192.168.230.131	80/tcp
VNC Server Unencrypted Data Transmission		4.8 (Medium)	70%	192.168.230.131	5900/tcp
Cleartext Transmission of Sensitive Information via HTTP		4.8 (Medium)	80%	192.168.230.131	80/tcp
Telnet Unencrypted Cleartext Login		4.8 (Medium)	70%	192.168.230.131	23/tcp
FTP Unencrypted Cleartext Login		4.8 (Medium)	70%	192.168.230.131	2121/tcp
FTP Unencrypted Cleartext Login		4.8 (Medium)	70%	192.168.230.131	21/tcp
SSH Weak Encryption Algorithms Supported		4.3 (Medium)	95%	192.168.230.131	22/tcp
SSL/TLS: Deprecated SSLv2 and SSLv3 Protocol Detection		4.3 (Medium)	98%	192.168.230.131	5432/tcp
SSL/TLS: Deprecated SSLv2 and SSLv3 Protocol Detection		4.3 (Medium)	98%	192.168.230.131	25/tcp
TWiki < 6.1.0 XSS Vulnerability		4.3 (Medium)	80%	192.168.230.131	80/tcp
SSL/TLS: RSA Temporary Key Handling 'RSA_EXPORT' Downgrade Issue (FREAK)		4.3 (Medium)	80%	192.168.230.131	25/tcp
phpMyAdmin 'error.php' Cross Site Scripting Vulnerability		4.3 (Medium)	99%	192.168.230.131	80/tcp
Apache HTTP Server 'httpOnly' Cookie Information Disclosure Vulnerability		4.3 (Medium)	99%	192.168.230.131	80/tcp
SSL/TLS: Certificate Signed Using A Weak Signature Algorithm		4.0 (Medium)	80%	192.168.230.131	5432/tcp
SSL/TLS: Certificate Signed Using A Weak Signature Algorithm		4.0 (Medium)	80%	192.168.230.131	25/tcp
SSL/TLS: Diffie-Hellman Key Exchange Insufficient DH Group Strength Vulnerability		4.0 (Medium)	80%	192.168.230.131	5432/tcp
SSL/TLS: Diffie-Hellman Key Exchange Insufficient DH Group Strength Vulnerability		4.0 (Medium)	80%	192.168.230.131	25/tcp
Tiki Wiki CMS Groupware XSS Vulnerability		3.5 (Low)	80%	192.168.230.131	80/tcp
TCP timestamps		2.6 (Low)	80%	192.168.230.131	general/tcp
SSH Weak MAC Algorithms Supported		2.6 (Low)	95%	192.168.230.131	22/tcp

(Applied filter:autofp=0 apply_overrides=1 notes=1 overrides=1 result_hosts_only=1 first=1 rows=100 sort-reverse=severity levels=hml min_qod=70) 1 - 56 of 56

5. Vulnerability Summary

Total Vulnerabilities Identified: 56

Severity Levels:

- Critical – Remote code execution vulnerabilities
- High – Authentication and service misconfigurations
- Medium – Information disclosure and outdated components
- Low – Security hardening recommendations

6. Major High-Risk Findings

• vsftpd 2.3.4 Backdoor Vulnerability

• UnrealIRCd Backdoor Command Execution

• Samba Remote Code Execution

• MySQL Weak Password Configuration

• Telnet Service Enabled (Unencrypted Login)

7. Risk Classification (CVSS Based)

Vulnerabilities were classified according to CVSS scoring methodology.

• Critical: Immediate remediation required

• High: Urgent patching recommended

• Medium: Scheduled remediation

• Low: Best practice improvements

8. Remediation Recommendations

Service Hardening:

• Disable Telnet and unused FTP services

• Restrict SMB exposure

Patch Management:

• Upgrade Apache, Samba, and MySQL

• Apply regular security updates

Credential Security:

• Remove default credentials

• Enforce strong password policies

Network Security:

• Configure firewall rules

• Limit exposed ports

• Implement IDS/IPS monitoring

9. Conclusion

Week 2 successfully demonstrated structured vulnerability assessment, risk classification, and professional documentation. The findings emphasize the importance of proactive vulnerability management and secure configuration practices.